Observability Analysis of PMSM

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***Abstract*—The current problems to successfully apply sensor-less control techniques for Permanent Magnet Synchronous Motor (PMSM) are the existence of operating conditions for which the observer performance is remarkably deteriorated. This is due to difficulties in estimating correctly the rotor position at low speed. The failure of sensor-less schemes in some particular operating conditions has been always recognized in experimental settings. However, there are no sufficient theoretical observability analysis for the PMSM. In the literature, only the sufficient observability condition has been presented. Therefore, the current work is aimed specially to necessary observability condition analysis. Thus, we give here the necessary and sufficient states observability condition for the PMSM. An example is presented to illustrate the results.**

***Index Terms*—PMSM, observability analysis, necessary and sufficient condition.**

I. Introduction

Synchronous Motors are widely used for low, medium and high power industrial applications. For high performance drives, it is possible to use various rotor configurations : rotor with permanent magnet, reluctance type and electrically exited rotors.

The PMSM is an ideal candidate for high performance industrial variable speed drives since it features simple structure, high energy efficiency, reliable operation and high power density. There are mainly too types of PMSMs: The PMSM with Surface mounted magnet (SPMSM) and the PMSM with Interior magnet (IPMSM). In the SPMSM, the magnets are located on the surface of the rotor and the machine behaves like a smooth air-grap machine (the direct and quadrature-axis synchronous inductances are equal, $L_d = L_q$) and there is only magnet torque produced. In the IPMSM, the magnets are inset or buried in the rotor ($L_d \neq L_q$) and both magnet torque and reluctance torque are produced.

Conventional PMSM controls are achieved by obtaining the rotor position and speed information through shaft sensors. However, the use of such sensors increases both the cost and the complexity and reduces the reliability and the robustness of the controlled drive system. For this reasons, an increasing interest is given to the sensor-less control which consists in replacing "hard" sensors with "soft" ones.

Since many excellent observers for PMSM sensor-less controls are available [1], [2], [3], [4], [5], [6], [7], the main research stream has been focused on searching for reliable speed and position estimation methods in the aim to replace the mechanical sensors with the observer in the control system [8], [9], [10], [11], [12]. However, The current problem to successfully apply sensor-lees controls for PMSM are the existence of operating regimes for which the observer performance is remarkably deteriorated due to the difficulties in estimating correctly motor speed and position.

The sensor-less techniques that have been proposed in the literature can be classified under two distinct categories, namely those that works at high speed and those that works at low speed. The high speed sensor-less techniques employed for PMSM motors are all directly or indirectly based on extracting position information from the motor BEMFs. Since the BEMFs are practically non-existent at low speed, these techniques can not operate in the low speed range [13]. The low speed sensor-less techniques are all based on the extracting of position information from a saliency based BEMF. Therefore, low speed sensor-less techniques are restricted to operate with PM motors that have inherent position dependant on saliencies. Such position-dependant saliencies are often characteristics of the construction of the motor, as is the case of the IPMSM. In the IPMSM, a saliency based BEMF can be generated even at low and zero speed by using various kinds of excitations [14], [15], [16], [17], [18].

The failure of sensor-less schemes in some particular operating conditions has been always recognized in experimental setting. In the case of Induction motors, the observability has been studied by many authors [19], [20]. However, there are no sufficient theoretical observability studies for the PMSM. Only the sufficient observability condition has been presented in literature. In [2], observability is analyzed in the case of constant high speed operation. In [21], the author give a sufficient observability condition (not necessary) of the PMSM in the particular case of constant speed operation. This work is aimed to the necessary and sufficient observability condition analysis. Thus, in this paper the observability is studied in the general case of variable speed operation. An example is presented to illustrate the results.

This paper is organized as follows: In the second section, the mathematical model of the PMSM is presented. In the third section, the non linear observability [22] is recalled. The observability analysis of the PMSM is given in section four. Finally, some concluding remarks are given in the last section.

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II. Mathematical Model Of the PMSM

A. IPMSM Model

The mathematical model of the IPMSM in the (d-q) rotating coordinate is given by equation (1) and (2) [23].

$$\begin{pmatrix} \frac{di_d}{dt} \\ \frac{di_q}{dt} \end{pmatrix} = \begin{pmatrix} -\frac{R}{L_d} & \frac{L_q}{L_d}P\omega \\ -\frac{L_d}{L_q}P\omega & -\frac{R}{L_q} \end{pmatrix} \begin{pmatrix} i_d \\ i_q \end{pmatrix} + \begin{pmatrix} \frac{1}{L_d} & 0 \\ 0 & \frac{1}{L_q} \end{pmatrix} \begin{pmatrix} v_d \\ v_q \end{pmatrix} + \begin{pmatrix} 0 \\ \frac{-P\varphi_m}{L_q}\omega \end{pmatrix} \quad (1)$$

$$\frac{d\omega}{dt} = \frac{P}{J}[(L_d - L_q)i_d + \phi_m]i_q - \frac{f_v}{J}\omega - \frac{T_l}{J} \quad (2)$$

where
ω is the rotor speed.
$\omega_e = P.\omega$ is the electric rotor speed.
R is the stator resistance.
L_d, L_q are the (d-q) stator inductance components.
P is the pair pole number.
J is the moment of inertia.
ϕ_m is the rotor flux.
f_v is the viscous friction.
T_l is the load torque.
$[i_d \quad i_q]^t$, $[v_d \quad v_q]^t$ are the (d-q) stator current and voltage vector respectively.
K_e is the BEMF constant.
Transforming the model given by (1) on the (α, β) fixed coordinate, (3) is derived [23]:

$$\begin{pmatrix} v_\alpha \\ v_\beta \end{pmatrix} = \begin{pmatrix} R + sL_\alpha & sL_{\alpha\beta} \\ sL_{\alpha\beta} & R + sL_\beta \end{pmatrix} \begin{pmatrix} i_\alpha \\ i_\beta \end{pmatrix} + \omega_e K_e \begin{pmatrix} -\sin(\theta_e) \\ \cos(\theta_e) \end{pmatrix} \quad (3)$$

where
$L_\alpha = L_0 + L_1 cos(2\theta_e)$
$L_\beta = L_0 - L_1 cos(2\theta_e)$
$L_{\alpha\beta} = L_1 \sin(2\theta_e)$
$L_0 = \frac{L_d + L_q}{2}$
$L_1 = \frac{L_d - L_q}{2}$
s is the differential operator.
The used inverse Park's transformation is:

$$\begin{pmatrix} x_\alpha \\ x_\beta \end{pmatrix} = \begin{pmatrix} \cos(\theta_e) & -\sin(\theta_e) \\ \sin(\theta_e) & \cos(\theta_e) \end{pmatrix} \begin{pmatrix} x_d \\ x_q \end{pmatrix} \quad (4)$$

With L_0 and L_1 terms, the equation system (3) becomes:

$$\begin{aligned} \begin{pmatrix} v_\alpha \\ v_\beta \end{pmatrix} &= R\begin{pmatrix} i_\alpha \\ i_\beta \end{pmatrix} + sL_0 \begin{pmatrix} i_\alpha \\ i_\beta \end{pmatrix} \\ &+ \omega_e K_e \begin{pmatrix} -\sin(\theta_e) \\ \cos(\theta_e) \end{pmatrix} \quad (5) \\ &+ L_1\omega_e \begin{pmatrix} -2\sin(2\theta_e) & 2\cos(2\theta_e) \\ 2\cos(2\theta_e) & 2\sin(2\theta_e) \end{pmatrix} \begin{pmatrix} i_\alpha \\ i_\beta \end{pmatrix} \\ &+ L_1 \begin{pmatrix} \cos(2\theta_e) & \sin(2\theta_e) \\ \sin(2\theta_e) & -\cos(2\theta_e) \end{pmatrix} s \begin{pmatrix} i_\alpha \\ i_\beta \end{pmatrix} \end{aligned}$$

Therefore, the dynamic model of IPMSM in the (α, β) fixed coordinate is given by equation (6) and (7):

$$\begin{aligned} \begin{pmatrix} \dot{i}_\alpha \\ \dot{i}_\beta \end{pmatrix} &= \Gamma^{-1} \Bigg[\begin{pmatrix} v_\alpha \\ v_\beta \end{pmatrix} - \\ &\begin{pmatrix} R - 2L_1\omega\sin(2\theta_e) & 2L_1\omega\cos(2\theta_e) \\ 2L_1\omega\cos(2\theta_e) & R + 2L_1\omega\sin(2\theta_e) \end{pmatrix} \begin{pmatrix} i_\alpha \\ i_\beta \end{pmatrix} \\ &-\omega K_e \begin{pmatrix} -\sin(\theta_e) \\ \cos(\theta_e) \end{pmatrix} \Bigg] \quad (6) \end{aligned}$$

$$\begin{aligned} \dot{\omega} &= \frac{P}{J}[2L_1(cos(\theta_e)i_\alpha + sin(\theta_e)i_\beta) + \phi_m](-sin(\theta_e)i_\alpha \\ &+ cos(\theta_e)i_\beta) - \frac{f_v}{J}\omega - \frac{T_l}{J} \quad (7) \end{aligned}$$

where

$$\Gamma^{-1} = \frac{1}{L_0^2 - L_1^2} \begin{pmatrix} L_0 - L_1\cos(2\theta_e) & -L_1 sin(2\theta_e) \\ -L_1 sin(2\theta_e) & L_0 + L_1\cos(2\theta_e) \end{pmatrix}$$

B. SPMSM Model

In the SPMSM we have $L_d = L_q$ then $L_1 = 0$.
Thus, from equation (6) and (7), we can deduce the dynamic model of the SPMSM:

$$\begin{pmatrix} \dot{i}_\alpha \\ \dot{i}_\beta \end{pmatrix} = \frac{1}{L_0} \Bigg[\begin{pmatrix} v_\alpha \\ v_\beta \end{pmatrix} - R\begin{pmatrix} i_\alpha \\ i_\beta \end{pmatrix} - \omega K_e \begin{pmatrix} -\sin(\theta_e) \\ \cos(\theta_e) \end{pmatrix} \Bigg] \quad (8)$$

$$\dot{\omega} = \frac{P}{J}\phi_m(-sin(\theta_e)i_\alpha + cos(\theta_e)i_\beta) - \frac{f_v}{J}\omega - \frac{T_l}{J} \quad (9)$$

III. Non Linear observability

In this section, the non linear observability [22] is recalled and an example is presented to illustrate that the rank criterion is only sufficient.
We consider systems of the form:

$$\sum : \begin{cases} \dot{x} = f(x, u) \\ y = h(x) \end{cases} \quad (10)$$

Where $x \in X \subset R^n$ is the state vector, $u \in U \subset R^m$ is the control vector, $y \in R^p$ is the output vector, f and h are C^∞ functions.
Definition 3-1(Indistinguishability and observability)
Consider the system $\sum$ and let x_0 and x_1 be two points of the state space X.
The pair x_0 and x_1 are indistinguishable (denoted $x_0 I x_1$) if ($\sum$,x_0) and ($\sum$,x_1) realize the same input-output map, i.e., for every admissible input $(u(t), [t_0 \quad t_1])$

$$\sum_{x_0}(u(t), [t_0, t_1]) = \sum_{x_1}(u(t), [t_0, t_1])$$

Indistinguishability I is an equivalence relation on X. $\sum$ is said to be observable at x_0 if $I(x_0) = x_0$ and $\sum$ is observable if $I(x) = x$ for every $x \in X$.
Definition 3-2(Locally weak observability)

Consider the system $\sum$ and let x_0 be a point of the state space X. $\sum$ is locally weakly observable at x_0 if there exist an open neighborhood V of x_0 such that for every open neighborhood v of x_0 contained in V, $I_v(x_0) = x_0$ and is locally weakly observable if it's so at every $x \in X$.

Definition 3-3(Rank Criterion)

A sufficient observability condition of the non linear system (10) is that the rank of the observability matrix J defined as:

$$J = \begin{pmatrix} dh \\ dL_f h \\ dL_f^2 h \\ . \\ . \\ . \\ dL_f^{n-1} h \end{pmatrix} \tag{11}$$

be equal to the state vector dimension n; $L_f^k h$ is the k^{th} order of Lie derivative of the function h with respect to the vector field f.

The following example illustrate that the Rank criterion is only sufficient.

Example : Consider the following system:

$$\begin{cases} \dot{x} = 1 \\ y = x^2 - x \end{cases} \tag{12}$$

System (12) can be written in the form of (10).

where

$$f(x) = 1$$

and

$$h(x) = x^2 - x$$

The order of this system is n=1. Applying the rank criterion to system (12) we obtain :

$$\begin{aligned} J &= \frac{\partial L_f^0 h(x)}{\partial x} \\ &= 2x - 1 \end{aligned} \tag{13}$$

From the expression of J, one remarks that for $x = \frac{1}{2}$ there are a singularity on observation. However, this do not means that system (12) is not observable at $x = \frac{1}{2}$ because it is possible to find observability using higher order Lie derivative of the output. In fact, let's look at the subspace generated from:

$$O = \begin{pmatrix} L_f^0 h \\ L_f^1 h \end{pmatrix} \tag{14}$$

The corresponding observability vector is:

$$\begin{aligned} J &= \begin{pmatrix} \frac{\partial L_f^0 h}{\partial x} \\ \frac{\partial L_f^1 h}{\partial x} \end{pmatrix} \\ &= \begin{pmatrix} 2x - 1 \\ 2 \end{pmatrix} \end{aligned} \tag{15}$$

For $x = \frac{1}{2}$ we obtain $J = \begin{pmatrix} 0 \\ 2 \end{pmatrix}$

Then

$$rank(J) = 1 = n$$

and the system (12) is observable.

Therefore, the Rank criterion is only sufficient.

IV. Observability Analysis Of the PMSM

To determine conditions under which it is possible to compute rotor speed and position information from measured output let's consider the state vector $x = [i_\alpha, \quad i_\beta, \quad \theta, \quad \omega]^t$ and the output vector $y = [i_\alpha, \quad i_\beta]^t$.

Voltages and currents are assumed to be measurable.

The order of the state vector of the PMSM is $n = 4$. Thus, according to the observability rank criterion mentioned earlier, the PMSM is observable if the following condition is fulfilled:

$$rank(J) = 4 \tag{16}$$

A. Observability analysis of the IPMSM

Consider the system (6) and (7) as:

$$\begin{pmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{x}_3 \\ \dot{x}_4 \end{pmatrix} = \frac{1}{L_0^2 - L_1^2} \begin{pmatrix} \Lambda_{11}\gamma_1 + \Lambda_{12}\gamma_2 \\ \Lambda_{21}\gamma_1 + \Lambda_{22}\gamma_2 \\ x_4 \\ T_e - m x_4 - \tau \end{pmatrix} \tag{17}$$

Where

$$\Lambda = \begin{pmatrix} L_0 - L_1 \cos(2\theta) & -L_1 sin(2\theta) \\ -L_1 sin(2\theta) & L_0 + L_1 \cos(2\theta) \end{pmatrix}$$

Λ_{ij} is the i^{th} row of the j^{th} column of the matrix Λ

$\gamma_1 = v_\alpha - (R - 2L_1 x_4 \sin(2x_3))x_1 + 2L_1 x_4 \cos(2x_3)x_2 + x_4 K_e sin(x_3)$

$\gamma_2 = v_\beta - (R - 2L_1 x_4 \sin(2x_3))x_2 - 2L_1 x_4 \cos(2x_3)x_1 - x_4 K_e cos(x_3)$

$T_e = \frac{P}{J}[2L_1(cos(x_3)x_1 + sin(x_3)x_2) + \phi_m](-sin(x_3)x_1 + cos(x_3)x_2)$ is the electromagnetic torque. Let

$$f(x) = \frac{1}{L_0^2 - L_1^2} \begin{pmatrix} \Lambda_{11}\gamma_1 + \Lambda_{12}\gamma_2 \\ \Lambda_{21}\gamma_1 + \Lambda_{22}\gamma_2 \\ x_4 \\ T_e - m x_4 - \tau \end{pmatrix}$$

and

$$h(x) = y$$

Then, look at the subspace generated from:

$$O_1 = \begin{pmatrix} h_1 \\ h_2 \\ L_f h_1 \\ L_f h_2 \end{pmatrix} \tag{18}$$

The associated observability matrix is:

$$J_1 = \frac{\partial}{\partial x} O_1 \quad (19)$$

$$J_1 = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ \frac{\partial L_f^1 h_1}{\partial x_1} & \frac{\partial L_f^1 h_1}{\partial x_2} & \frac{\partial L_f^1 h_1}{\partial x_3} & \frac{\partial L_f^1 h_1}{\partial x_4} \\ \frac{\partial L_f^1 h_2}{\partial x_1} & \frac{\partial L_f^1 h_2}{\partial x_2} & \frac{\partial L_f^1 h_2}{\partial x_3} & \frac{\partial L_f^1 h_2}{\partial x_4} \end{pmatrix} \quad (20)$$

Computing the corresponding determinant gives:

$$\Delta_1 = \frac{\partial L_f^1 h_1}{\partial x_3} . \frac{\partial L_f^1 h_2}{\partial x_4} - \frac{\partial L_f^1 h_2}{\partial x_3} . \frac{\partial L_f^1 h_1}{\partial x_4} \quad (21)$$

$$\begin{aligned} \Delta_1 &= [[2L_1 \sin(2x_3)\gamma_1 + (L_0 - L_1 \cos(2x_3))\frac{\partial \gamma_1}{\partial x_3} \\ &- 2L_1 \cos(2x_3)\gamma_2 - L_1 \sin(2x_3)\frac{\partial \gamma_2}{\partial x_3}].[-L_1 \sin(2x_3)\frac{\partial \gamma_1}{\partial x_4} \\ &+ (L_0 + L_1 \cos(2x_3))\frac{\partial \gamma_2}{\partial x_4}] - [-2L_1 \cos(2x_3)\gamma_1 \\ &- L_1 sin(2x_3)\frac{\partial \gamma_1}{\partial x_3} - 2L_1 \sin(2x_3)\gamma_2 \\ &+ (L_0 - L_1 \cos(2x_3))\frac{\partial \gamma_2}{\partial x_3}][(L_0 - L_1 \cos(2x_3))\frac{\partial \gamma_1}{\partial x_4} \\ &- L_1 \sin(2x_3)\frac{\partial \gamma_2}{\partial x_4}]]/(L_0^2 - L_1^2) \end{aligned} \quad (22)$$

Where

$\frac{\partial \gamma_1}{\partial x_3} = 4L_1 x_4 \cos(2x_3)x_1 - 4L_1 x_4 \sin(2x_3)x_2 + x_4 K_e \cos(x_3)$
$\frac{\partial \gamma_2}{\partial x_3} = 4L_1 x_4 \sin(2x_3)x_1 - 4L_1 x_4 \cos(2x_3)x_2 + x_4 K_e \sin(x_3)$
$\frac{\partial \gamma_1}{\partial x_4} = 2L_1 \sin(2x_3)x_1 + 2L_1 \cos(2x_3)x_2 + K_e \sin(x_3)$
$\frac{\partial \gamma_2}{\partial x_4} = -2L_1 \cos(2x_3)x_1 - 2L_1 \sin(2x_3)x_2 - K_e \cos(x_3)$

In this case of IPMSM ($L_1 \neq 0$), at zero speed ($x_4 = 0$) the determinant has the following expression:

$$\begin{aligned} \Delta_1 &= [[2L_1 \sin(2x_3)(v_\alpha - Rx_1) - 2L_1 \cos(2x_3)(v_\beta \\ &- Rx_2)].[-L_1 \sin(2x_3)(2L_1 \sin(2x_3)x_1 \\ &+ 2L_1 \cos(2x_3)x_2 + K_e \sin(x_3)) + (L_0 \\ &+ L_1 \cos(2x_3))(-2L_1 \cos(2x_3)x_1 \\ &- 2L_1 \sin(2x_3)x_2 - K_e \cos(x_3))]]/(L_0^2 - L_1^2) \end{aligned} \quad (23)$$

Remark 4-1
Looking at the previous expression (23), we remark that at zero speed operation Δ_1 depends on current and voltage. Therefore, we have always the opportunity to find again the observability propriety by injection of a continue current. Thus, we can conclude that the IPMSM is always observable.

B. Observability analysis of the SPMSM

In this section, we present the observability analysis of the SPMSM and we give a sufficient condition of loss of the observability property.

In this case, we have $L_d = L_q$ then $L_1 = 0$. Therefore, the expression of determinant given in (22) becomes:

$$\Delta_1 = -K_e^2 x_4 \quad (24)$$

Remark 4-2 The determinant Δ_1 is dependant only on x_4. Thus, any input $v_{\alpha,\beta}$ results in a locally weakly observable system over the subspace $\Omega = \{x : x_4 \neq 0\}$. Then the previous condition indicates that a sufficient observability condition can be fulfilled if the motor operates away from zero speed condition. However, $x_4 = 0$ is not a sufficient condition for concluding that the SPMSM is not observable.

Now the question is to look if higher derivatives of the output overcome the observability singularity at zero speed. For that, let's consider the model of SPMSM (8) in the form of (10) where:

$$f(x,u) = \begin{pmatrix} ax_1 + bx_4 sin(x_3) + cv_\alpha \\ ax_2 - bx_4 cos(x_3) + cv_\beta \\ x_4 \\ k_t(-sin(x_3)x_1 + cos(x_3)x_2) - mx_4 - \tau \end{pmatrix}$$

and $h(x) = [x_1, \quad x_2]^t$, with $a = \frac{-R}{L_0}$, $b = \frac{K_e}{L_0}$, $c = \frac{1}{L_0}$, $k_t = \frac{p\phi_m}{J}$, $m = \frac{f_v}{J}$ and $\tau = \frac{T_l}{J}$

Let's look to the following subspace generated from:

$$O_2 = \begin{pmatrix} h_1 \\ h_2 \\ L_f h_1 \\ L_f h_2 \\ L_f^2 h_1 \\ L_f^2 h_2 \end{pmatrix} \quad (25)$$

The associated observability matrix is:

$$J_2 = \frac{\partial}{\partial x} O_2 \quad (26)$$

Condition (16) can be tested by searching for a regular matrix constructed from any four rows of matrix J_2. Let's consider the 1^{st}, 2^{nd}, 5^{th} and the 6^{th} rows of J_2.

$$J_2 = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ \frac{\partial L_f^2 h_1}{\partial x_1} & \frac{\partial L_f^2 h_1}{\partial x_2} & \frac{\partial L_f^2 h_1}{\partial x_3} & \frac{\partial L_f^2 h_1}{\partial x_4} \\ \frac{\partial L_f^2 h_2}{\partial x_1} & \frac{\partial L_f^2 h_2}{\partial x_2} & \frac{\partial L_f^2 h_2}{\partial x_3} & \frac{\partial L_f^2 h_2}{\partial x_4} \end{pmatrix} \quad (27)$$

Computing the corresponding determinant gives:

$$\begin{aligned} \Delta_2 &= \frac{\partial L_f^2 h_1}{\partial x_3} . \frac{\partial L_f^2 h_2}{\partial x_4} - \frac{\partial L_f^2 h_2}{\partial x_3} . \frac{\partial L_f^2 h_1}{\partial x_4} \\ &= b^2[-a^2 + am + 2k_t(-cos(x_3)x_1 \\ &- sin(x_3)x_2)]x_4 - 2b^2 x_4^3 - b^2(a - m)\dot{x}_4 \end{aligned} \quad (28)$$

Remark 4-3 The expression of J_2 depends not only on x_4 but also on $\dot{x}_4$. Thus, any input $v_{\alpha,\beta}$ results in a locally weakly observable system over the subspace $\Omega = \{x : x_4 \neq 0 \quad Or \quad \dot{x}_4 \neq 0\}$.
This proves that a sufficient observability condition can be obtained at zero speed operation as long as the motor operates away from constant speed condition.
We have proved that the SPMSM is locally weakly observable when $\dot{x}_4 \neq 0$ or $x_4 \neq 0$. Now, the problem is if $\dot{x}_4 = 0$, is it possible to find observability at zero speed operation using higher order derivatives of the output? For that let's consider the model of the SPMSM in the particular case of constant speed ($\dot{x}_4 = 0$). In this case, the function $f(x,u)$ is replaced by

$$f_0(x,u) = \begin{pmatrix} ax_1 + bx_4 sin(x_3) + cu_\alpha \\ ax_2 - bx_4 cos(x_3) + cu_\beta \\ x_4 \\ 0 \end{pmatrix}$$

Then look now at the subspace generated from:

$$O_3 = \begin{pmatrix} h_1 \\ h_2 \\ L_f h_1 \\ L_f h_2 \\ L_f^2 h_1 \\ L_f^2 h_2 \\ L_f^3 h_1 \\ L_f^3 h_2 \\ L_f^4 h_1 \\ L_f^4 h_2 \end{pmatrix} \tag{29}$$

The associated observability matrix is:

$$J_3 = \frac{\partial}{\partial x} O_3 = \begin{pmatrix} a_1 & b_1 & c_1 & d_1 \\ a_2 & b_2 & c_2 & d_2 \\ a_3 & b_3 & c_3 & d_3 \\ a_4 & b_4 & c_4 & d_4 \\ a_5 & b_5 & c_5 & d_5 \\ a_6 & b_6 & c_6 & d_6 \\ a_7 & b_7 & c_7 & d_7 \\ a_8 & b_8 & c_8 & d_8 \\ a_9 & b_9 & c_9 & d_9 \\ a_{10} & b_{10} & c_{10} & d_{10} \end{pmatrix} \tag{30}$$

With
$a_1 = 1, \quad b_1 = 0, \quad c_1 = 0, \quad d_1 = 0$
$a_2 = 0, \quad b_2 = 1, \quad c_2 = 0, \quad d_2 = 0$
$a_3 = a, \quad b_3 = 0, \quad c_3 = bx_4 cos(x_3), \quad d_3 = bsin(x_3)$
$a_4 = 0, \quad b_4 = a, \quad c_4 = bx_4 sin(x_3), \quad d_4 = -bcos(x_3)$
$a_5 = a^2, \quad b_5 = 0, \quad c_5 = abx_4 cos(x_3) - bx_4^2 sin(x_3), \quad d_5 = absin(x_3) + 2bx_4 cos(x_3)$
$a_6 = 0, \quad b_6 = a^2, \quad c_6 = abx_4 sin(x_3) + bx_4^2 cos(x_3), \quad d_6 = -abcos(x_3) + 2bx_4 sin(x_3)$
$a_7 = a^3, \quad b_7 = 0, \quad c_7 = a^2 bcos(x_3) x_4 + (-absin(x_3) x_4 - bx_4^2 cos(x_3)) x_4, \quad d_7 = a^2 bsin(x_3) + abx_4 cos(x_3) - bx_4^2 sin(x_3) + (abcos(x_3) - 2bx_4 sin(x_3)) x_4$
$a_8 = 0, \quad b_8 = a^3, \quad c_8 = a^2 bsin(x_3) x_4 + (abcos(x_3) x_4 - bx_4^2 sin(x_3)) x_4, \quad d_8 = -a^2 bcos(x_3) + abx_4 sin(x_3) + bx_4^2 cos(x_3) + (absin(x_3) + 2bx_4 cos(x_3)) x_4$
$a_9 = a^4, \quad b_9 = 0, \quad c_9 = a^3 bx_4 cos(x_3) + [-a^2 bx_4 sin(x_3) + (-abx_4 cos(x_3) + bx_4^2 sin(x_3)) x_4] x_4, \quad d_9 = a^3 bsin(x_3) + 2a^2 bx_4 cos(x_3) - 3abx_4^2 sin(x_3) - 4bx_4^3 cos(x_3)$
$a_{10} = 0, \quad b_{10} = a^4, \quad c_{10} = a^3 bx_4 sin(x_3) + [a^2 bx_4 cos(x_3) + (-abx_4 sin(x_3) - bx_4^2 cos(x_3)) x_4] x_4, \quad d_{10} = -a^3 bcos(x_3) + 2a^2 bx_4 sin(x_3) + 3abx_4^2 cos(x_3) - 4bx_4^3 sin(x_3)$

In this case (constant speed) it is obvious that the SPMSM is locally weakly observable if $x_4 \neq 0$. However, if we consider also $x_4 = 0$ (this correspond to zero speed and acceleration) then, in this case, the observability matrix J_3 becomes:

$$J_3 = \frac{\partial}{\partial x} O_3 = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ a & 0 & 0 & bsin(x_3) \\ 0 & a & 0 & -bcos(x_3) \\ a^2 & 0 & 0 & absin(x_3) \\ 0 & a_2 & 0 & -abcos(x_3) \\ a^3 & 0 & 0 & a^2 bsin(x_3) \\ 0 & a^3 & 0 & -a^2 bcos(x_3) \\ a^4 & 0 & 0 & a^3 bsin(x_3) \\ 0 & a^4 & 0 & -a^3 bcos(x_3) \end{pmatrix} \tag{31}$$

Remark 4-4 From (31) a recurrence equation can be obtained and can be generalized for higher derivatives.:

$$\frac{\partial}{\partial x} L_f^k h_i = a \frac{\partial}{\partial x} L_f^{k-1} h_i; \quad k = 2...4, \quad i = 1,2. \tag{32}$$

Thus, it is not necessary to compute Lie derivative superior than four because we will always find the same relation (32). So we can conclude a sufficient condition of loss of observability of the SPMSM over the subspace:

$$\Omega = \{x : x_4 = 0 \quad and \quad \dot{x}_4 = 0\} \tag{33}$$

Example [24] Figure 1. presents the response of a SPMSM using a Super Twisting Observer to estimate the rotor speed and position. In this figure, we illustrate the unobservability property of the SPMSM at zero speed and acceleration.
A physically interpretation is related to the fact that the rotor position information is extracted only from BEMFs. Since the BEMFs are practically non existent at zero speed and acceleration, then, it is impossible to have any information about rotor position. Therefore, the SPMSM lose observability in these conditions.

V. Conclusion

In this paper, the observability analysis of the PMSM has been presented to determine possibilities and limitations to achieve reliable sensor-less control schemes. In this study, a necessary and sufficient observability condition has been presented for the PMSM.

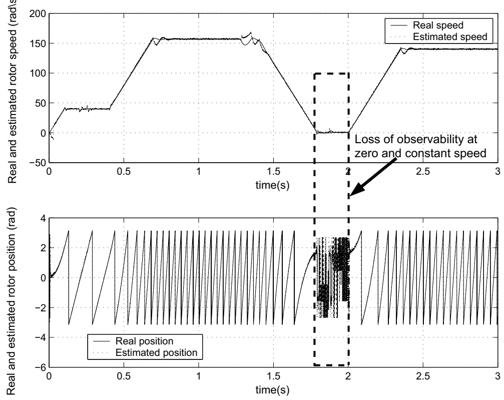

Fig. 1. Loss of observability at zero and constant speed operation

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